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United States Patent	12406949
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Vincent; Michael B.

Semiconductor device with RF interposer and method therefor

Abstract

A method of forming a self-aligned waveguide is provided. The method includes providing a radio frequency (RF) interposer. The RF interposer includes a non-conductive substrate, a radiating element formed on the non-conductive substrate, and a cavity formed in the non-conductive substrate. A packaged semiconductor die is affixed in the cavity of the RF interposer. A conductive material is dispensed to form a conductive path between a conductive connector of the packaged semiconductor die and the radiating element.

Inventors:	Vincent; Michael B. (Chandler, AZ)
Applicant:	NXP USA, INC. (Austin, TX)
Family ID:	85153043
Assignee:	NXP USA, INC. (Austin, TX)
Appl. No.:	17/393954
Filed:	August 04, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230044903 A1	Feb. 09, 2023

Publication Classification

Int. Cl.: H01L23/66 (20060101); H01L21/50 (20060101); H01L23/14 (20060101); H01Q1/22 (20060101)

U.S. Cl.:

CPC H01L23/66 (20130101); H01L21/50 (20130101); H01L23/14 (20130101); H01Q1/2283

Field of Classification Search

CPC: H01L (23/66); H01L (21/50); H01L (23/14); H01L (23/3107); H01L (23/3135); H01L (24/16); H01L (24/42); H01L (2223/6605); H01L (2223/6677); H01L (23/13); H01L (23/49816); H01L (23/49827); H01Q (1/2283); H01Q (1/521); H01Q (7/00); H01Q (21/0037)

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Primary Examiner: Gheyas; Syed I

Background/Summary

BACKGROUND

Field

(1) This disclosure relates generally to semiconductor device packaging, and more specifically, to a semiconductor device with radio frequency (RF) interposer and method of forming the same.

Related Art

(2) Today, there is an increasing trend to include radar systems in vehicles such as automobiles, trucks, buses, and the like in order to provide a driver with enhanced awareness of objects around the driver's vehicle. As the vehicle approaches objects (e.g. other cars, pedestrians, and obstacles) or as objects approach the vehicle, a driver cannot always detect the object and perform intervention actions needed to avoid a collision with the object. An automotive radar system mounted on a vehicle can detect the presence of objects including other vehicles in proximity to the vehicle and provide the driver with timely information so that the driver can perform possible intervention actions. However, such automotive radar system can significantly impact the cost of the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention is illustrated by way of example and is not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

(2) FIG. 1 illustrates, in a simplified top-side-up plan view, an example semiconductor device having an RF interposer at a stage of manufacture in accordance with an embodiment.

(3) FIG. 2 illustrates, in a simplified bottom-side-up plan view, the example semiconductor device in accordance with an embodiment.

(4) FIG. 3 through FIG. 7 illustrate, in simplified cross-sectional views, the example semiconductor device at stages of manufacture in accordance with an embodiment.

(5) FIG. 8 illustrates, in a simplified bottom-side-up plan view, an alternative example semiconductor device having an RF interposer in accordance with an embodiment.

(6) FIG. 9 illustrates, in a simplified top-side-up plan view, the alternative example semiconductor device in accordance with an embodiment.

(7) FIG. 10 illustrates, in a simplified cross-sectional view, the alternative example semiconductor device at a stage of manufacture in accordance with an embodiment.

DETAILED DESCRIPTION

(8) Generally, there is provided, a semiconductor device with an RF interposer. The semiconductor device includes a singulated packaged semiconductor die affixed in a cavity of the RF interposer. The packaged semiconductor die is formed as an encapsulated semiconductor die including a plurality of conductive connectors. The RF interposer includes a non-conductive substrate having radiating elements and other conductive and interconnecting features. The radiating elements may be configured for propagation of RF signals such as radar, Wi-Fi, UWB, 5G and 6G signals, for example. The cavity is formed in a bottom side of the substrate. With the packaged semiconductor die affixed in the cavity, conductive material is applied to form conductive interconnects between conductive pads of the RF interposer substrate and respective conductive connectors of the packaged semiconductor die. The conductive material may be applied by dispensing, jetting, and the like. A reflow operation causes the respective conductive connectors to wet the conductive material of the conductive interconnects forming conductive paths between the packaged semiconductor die and the radiating elements, for example. By forming the conductive interconnects after the packaged semiconductor die and RF interposer are joined together, supply

chain complexity and costs may be reduced.

(9) FIG. 1 illustrates, in a simplified top-side-up plan view, an example semiconductor device **100** having an RF interposer **102** at a stage of manufacture in accordance with an embodiment. In this embodiment, the semiconductor device **100** includes an RF interposer **102** and packaged semiconductor die located in a cavity **122**. The cavity **122**, shown as a dashed outline for reference, is formed at a bottom side of the RF interposer **102**. In this embodiment, the semiconductor device **100** may be characterized as an upward launching device configured for launching radio frequency (RF) signals in an upward direction, for example. Simplified cross-sectional views of the example semiconductor device **100** taken along line A-A at stages of manufacture are depicted in FIG. 3 through FIG. 7.

(10) The RF interposer **102** includes radiating element (e.g., signal launcher, antenna) structures **124** and **126** formed on a non-conductive substrate **104**. The substrate **104** may be formed from suitable non-conductive materials such as laminate (e.g., FR4), ceramic, glass, and the like. In this embodiment, the radiating element structures **124** and **126** are configured for propagation of RF signals such as radar, Wi-Fi, UWB, 5G and 6G signals. The group of radiating element structures **124** may be characterized as transmitter (TX) radiating element structures and the group of radiating element structures **126** may be characterized as receiver (RX) radiating element structures, for example. Each of the TX radiating element structures **124** include a conductive (e.g., metal) radiating element **106** and conductive (e.g., metal) surrounding ring **110** and each of the RX radiating element structures **124** include a radiating element **108** and surrounding ring **112**. Conductive pads **114** and **116** are connected to respective radiating elements **106** and **108** by way of conductive (e.g., metal) traces formed at the top side of the substrate **104**. Likewise, conductive pads **118** and **120** are connected to surrounding rings **110** and **112** by way of conductive traces formed at the top side of the substrate **104**. In this embodiment, conductive pads **114-120** are interconnected to corresponding conductive pads formed at the bottom side of the substrate **104** by way of conductive vias.

(11) FIG. 2 illustrates, in a simplified bottom-side-up plan view, the example semiconductor device **100** at the stage of manufacture of FIG. 1 in accordance with an embodiment. At this stage, the semiconductor device **100** includes packaged semiconductor die **202** affixed in the cavity **122** of the RF interposer **102** by way of an adhesive material **208**. The adhesive material **208** may be applied in the form of a die attach film (DAF) or may be dispensed, jetted, or printed, for example. In this embodiment, the packaged semiconductor die **202** is depicted as an example wafer-level packaging (WLP) package type having a plurality of conductive connectors (e.g., solder balls) **210** and **212**. In other embodiments, other package configuration types such as chip-scale packaging (CSP) and the like are anticipated by this disclosure.

(12) The packaged semiconductor die **202** includes a semiconductor die **204** encapsulated on five sides (e.g., four sidewalls side plus backside) with an encapsulant (e.g., epoxy) **206**. The conductive connectors **210** and **212** are affixed at the bottom (e.g., active) side of the packaged semiconductor die **202** and conductively connected to semiconductor die **204** by way of a redistribution layer (not shown), for example.

(13) The semiconductor die **204** has an active side (e.g., major side having circuitry) and a backside (e.g., major side opposite of the active side). The semiconductor die **204** includes bond pads (not shown) at the active side configured for connection to the conductive connectors **210** and **212** by way of the redistribution layer, for example. In this embodiment, semiconductor die **204** is configured in an active-side-down orientation with the active side exposed at the bottom side of the encapsulant **206**. The semiconductor die **204** may be formed from any suitable semiconductor material, such as silicon, germanium, gallium arsenide, gallium nitride, and the like. The semiconductor die **204** further includes any digital circuits, analog circuits, RF circuits, memory, processor, the like, and combinations thereof at the active side.

(14) In this embodiment, the RF interposer **102** further includes bottom conductive (e.g., metal)

pads **214-220** formed on an opposite side of the substrate **104** from respective conductive pads **114-120**. The bottom conductive pads **214-220** are interconnected to the conductive pads **114-120** by way of conductive vias, for example. In this embodiment, the bottom conductive pads **214** are interconnected to respective radiating elements **106** of the TX radiating element structures **124** and conductive pad **218** is interconnected to surrounding rings **110** of the TX radiating element structures **124**. Likewise, the bottom conductive connectors **216** are interconnected to respective radiating elements **108** of the RX radiating element structures **126** and conductive connector **220** is interconnected to surrounding rings **112** of the RX radiating element structures **124**.

(15) At this stage, interconnects **222-228** are applied between the bottom conductive pads **214-220** and respective conductive connectors of the packaged semiconductor die **202** by way of an applied (e.g., dispensed, jetted, sputtered, stenciled, printed, placed) conductive material to form conductive paths between the packaged semiconductor die **202** and the radiating elements **106** and **108** of the TX and RX radiating element structures **124** and **126** of the RF interposer **102**. Similarly, the interconnects **226** and **228** are configured to provide a ground voltage supply connection to the surrounding rings **110** and **112** of the RX and TX radiating element structures **124** and **126** in this embodiment.

(16) FIG. 3 through FIG. 7 illustrate, in simplified cross-sectional views, the example semiconductor device **100** taken along line A-A of FIG. 1 at stages of manufacture in accordance with an embodiment. In the embodiments of FIG. 3 through FIG. 7, the packaged semiconductor die **202** is depicted as a WLP package type having conductive connectors **210** and **212** interconnected with the semiconductor die **204** by way of a redistribution layer (not shown). Even though the embodiments depicted in FIG. 3 through FIG. 7 exemplify WLP type package configurations, embodiments in other package configuration types (e.g., ball grid array, chip scale packaging) are anticipated by this disclosure.

(17) FIG. 3 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device **100** taken along line A-A at a stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **100** includes the packaged semiconductor die **202** positioned over the cavity **122** of the RF interposer **102** to form an assembly by way of the adhesive material **208**. In one embodiment, the assembly is formed by applying the adhesive material **208** to surfaces of the cavity **122** before the packaged semiconductor die **202** and the RF interposer **102** are joined together. In another embodiment, the assembly is formed by applying the adhesive material **208** to surfaces of the packaged semiconductor die **202** before the packaged semiconductor die **202** is joined with the RF interposer **102**. The adhesive material **208** may be applied in the form of a DAF or may be dispensed, jetted, or printed, for example. It may be desirable for the adhesive material **208** to fill the vertical space between sidewalls of the packaged semiconductor die **202** and RF interposer **102** such that the adhesive material **208** is substantially coplanar at the bottom surface. This allows for a smooth (e.g., planar) transition from the packaged semiconductor die **202** to the RF interposer **102** when applying conductive material to form interconnects **222-228** between the bottom conductive pads **214-220** and respective conductive connectors, for example.

(18) In this embodiment, the RF interposer further includes conductive (e.g., metal) signal reflectors **302** and **304** interconnected with respective surrounding rings **110** and **112** by way of conductive vias **308**. The conductive vias **308** are configured and arranged to form a vertical conductive structure (e.g., wall, fence) substantially surrounding each of the respective radiating elements **106** and **108**. The bottom conductive pads **214** and **216** are interconnected to respective radiating elements (e.g., signal launcher, antenna) **106** and **108** by way of conductive vias **306**. In this embodiment, the signal reflectors **302** and **304** are located on a major side of the substrate **104** opposite of the respective radiating elements **106** and **108**.

(19) FIG. 4 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance

with an embodiment. At this stage, the interconnects **222** and **224** are applied, by way of an application apparatus **402**, on bottom portions of the assembly **404** formed with the packaged semiconductor die **202** and the RF interposer **102**. The interconnects **222** and **224** are formed by way of applying a conductive material (e.g., silver ink, metal particle filled epoxy, sputtered metals such as titanium, titanium-tungsten, copper, and the like) to form conductive paths between the bottom conductive pads **214** and **216** and respective signal conductive connectors **212** of the packaged semiconductor die **202**. In this embodiment, the interconnects **222** and **224** are formed by applying the conductive material in a manner that substantially covers the exposed bottom conductive pads **214** and **216** and extends inward to substantially about an edge of conductive connectors **212**. The conductive material may be applied by way of application apparatus **402** and suitable techniques such as dispensing, jetting, sputtering, stenciling, printing, placing, and the like, for example. It may be desirable for the conductive material to be characterized as reflowable and/or solderable material, for example.

(20) FIG. 5 illustrates, in a simplified top-side-up cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the assembly **404** including interconnects **222** and **224** is flipped vertically and positioned over a printed circuit board (PCB) **502** for mounting to the PCB. In this embodiment, the assembly **404** formed with the packaged semiconductor die **202** and the RF interposer **102** is positioned in a top-side-up orientation over the PCB **502** with the conductive connectors **210** aligned with corresponding PCB pads **504** located at a top side of the PCB. In some embodiments, a solder paste or flux **506** may be applied onto the PCB pads **504** to enhance solder wetting during a reflow operation. The solder paste **506** or flux may be stencil printed onto the PCB pads **504**, for example. In this embodiment, the solder paste **506** includes a combination of solder particles and a solder flux material. The solder particles may include a mixture of tin, copper, and other metals, for example.

(21) FIG. 6 illustrates, in a simplified top-side-up cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the assembly **404** is affixed to the PCB **502** by way of the conductive connectors **210**. In this embodiment, the assembly **404** including interconnects **222** and **224** is subjected to a reflow operation. The reflow operation causes the conductive connectors **210** (and solder paste **506**) to wet the PCB pads **504** and form conductive connections while establishing a physical (e.g., structural) attachment to the PCB **502**. The reflow operation also causes the conductive connectors **212** to wet the conductive material of the interconnects **222** and **224** and form the conductive connections between the semiconductor die **204** and the bottom conductive pads **214** and **216** thus interconnecting the packaged semiconductor die **202** with the radiating elements **106** and **108**.

(22) FIG. 7 illustrates, in a simplified top-side-up cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, a waveguide structure **702** is aligned and affixed to the assembly **404**. In this embodiment, an interface material **710** is disposed between the RF interposer **102** and the waveguide structure **702**. The interface material **710** may serve as an adhesive for affixing the waveguide structure **702** to the RF interposer **102** and/or may serve as a stress buffer material for minimizing stress between the waveguide structure **702** and the RF interposer **102**, for example. In some embodiments, the interface material **710** may be conductive and patterned as a form of shielding to prevent RF signal radiation from propagating laterally from the intended vertical path.

(23) In this embodiment, the waveguide structure **702** includes waveguides **706** formed in a waveguide body **704**. Waveguide antennas **708** are located at the top of the waveguides **706** at the top side of the waveguide structure **702**. In this embodiment, the waveguide structure **702** may be characterized as an air cavity waveguide antenna. For example, the waveguide body **704** may be formed from a metal, plastic, or laminate material with the waveguides **706** formed as air cavities.

The waveguides **706** may be formed with a metal lining when the waveguide body **704** is formed from a plastic or laminate material, for example. Alignment features (not shown) may be incorporated in the waveguide structure **702** and the RF interposer **102** to facilitate proper alignment of the waveguides **706** with respective radiating elements **106** and **108**.

(24) FIG. **8** illustrates, in a simplified bottom-side-up plan view, an alternative example semiconductor device **800** having an RF interposer **802** at a stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **800** includes a packaged semiconductor die **822** affixed in a cavity **842** of an RF interposer **802**. In this embodiment, the semiconductor device **800** may be characterized as downward launching device configured for launching radio frequency (RF) signals in a downward direction, for example. A simplified cross-sectional view of the example semiconductor device **800** taken along line B-B at a stage of manufacture is depicted in FIG. **10**.

(25) The packaged semiconductor die **822** includes a semiconductor die **824** encapsulated on five sides (e.g., four sidewalls side plus backside) with an encapsulant (e.g., epoxy) **826**. The conductive connectors **830** and **832** are affixed at the bottom (e.g., active) side of the packaged semiconductor die **822** and conductively connected to semiconductor die **824** by way of a redistribution layer (not shown), for example.

(26) The semiconductor die **824** has an active side (e.g., major side having circuitry) and a backside (e.g., major side opposite of the active side). The semiconductor die **824** includes bond pads (not shown) at the active side configured for connection to the conductive connectors **830** and **832** by way of the redistribution layer, for example. In this embodiment, semiconductor die **824** is configured in an active-side-down orientation with the active side exposed at the bottom side of the encapsulant **826**. The semiconductor die **824** may be formed from any suitable semiconductor material, such as silicon, germanium, gallium arsenide, gallium nitride, and the like. The semiconductor die **824** further include any digital circuits, analog circuits, RF circuits, memory, processor, the like, and combinations thereof at the active side.

(27) The RF interposer **802** includes radiating element (e.g., signal launcher, antenna) structures **844** and **846** formed on a non-conductive substrate **804**. The substrate **804** may be formed from suitable non-conductive materials such as laminate (e.g., FR4), ceramic, glass, and the like. In this embodiment, the radiating element structures **844** and **846** are configured for propagation of RF signals such as radar, Wi-Fi, UWB, 5G and 6G signals. The group of radiating element structures **844** may be characterized as transmitter (TX) radiating element structures and the group of radiating element structures **846** may be characterized as receiver (RX) radiating element structures, for example. Each of the TX radiating element structures **844** include a conductive (e.g., metal) radiating element **806** and conductive (e.g., metal) surrounding ring **810** and each of the RX radiating element structures **844** include a radiating element **808** and surrounding ring **812**.

Conductive pads **814** and **816** are connected to respective radiating elements **806** and **808** by way of conductive (e.g., metal) traces formed at the bottom side of the substrate **804**. Likewise, conductive pads **818** and **820** are connected to surrounding rings **810** and **812** by way of conductive traces formed at the bottom side of the substrate **804**.

(28) The RF interposer **802** further includes the cavity **842** formed in the substrate **804** at a bottom side of the interposer. The packaged semiconductor die **822** is affixed in the cavity **842** by way of an adhesive material **828**. The adhesive material **828** may be applied to surfaces of the cavity **842** before the packaged semiconductor die **822** and the RF interposer **802** are joined together, or the adhesive material **828** may be applied to surfaces of the packaged semiconductor die **822** before the packaged semiconductor die **822** is joined with the RF interposer **802**. The adhesive material **828** may be applied in the form of a die attach film (DAF) or may be dispensed, jetted, or printed, for example. It may be desirable for the adhesive material **828** to fill the vertical space between sidewalls of the packaged semiconductor die **822** and RF interposer **802** such that the adhesive material **828** is substantially coplanar at the bottom surface. This allows for a smooth (e.g., planar)

transition from the packaged semiconductor die **822** to the RF interposer **802** when forming interconnects **834-840**, for example.

(29) At this stage, interconnects **834-840** are applied between the conductive pads **814-820** and respective conductive connectors of the packaged semiconductor die **822** by way of an applied (e.g., dispensed, jetted, sputtered, stenciled, printed, placed) conductive material. In this embodiment, the interconnects **834-840** are formed by applying the conductive material in a manner that substantially covers the exposed bottom conductive pads **814-820** and extends inward to substantially abut an edge of conductive connectors **832**. It may be desirable for the conductive material to be characterized as reflowable and/or solderable material, for example. The conductive material is applied to form conductive paths between the packaged semiconductor die **822** and the radiating element structures **844** and **846** of the RF interposer **802**. For example, the interconnects **838** and **840** are configured to provide a ground voltage supply connection to the surrounding rings **810** and **812** of the TX and RX radiating element structures **844** and **846** in this embodiment.

(30) FIG. **9** illustrates, in a simplified top-side-up plan view, the example semiconductor device **800** at the stage of manufacture of FIG. **8** in accordance with an embodiment. In this embodiment, a top side of the RF interposer **802** is depicted. For reference, the cavity **842** formed at the bottom side of the RF interposer **802** is shown as a dashed outline. In this embodiment, the RF interposer **802** further includes a plurality of conductive (e.g., metal) signal reflectors **902** and **904**. The signal reflectors **902** and **904** are interconnected with respective surrounding rings **810** and **812** by way of conductive vias (not shown), for example.

(31) FIG. **10** illustrates, in a simplified top-side-up cross-sectional view, the example semiconductor device **800** taken along line B-B at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **800** includes an assembly formed with the packaged semiconductor die **822** and the RF interposer **802** affixed on a PCB **1002** by way of the conductive connectors **830**. In this embodiment, the assembly including interconnects **834** and **836** is subjected to a reflow operation. The reflow operation causes the conductive connectors **830** to wet PCB pads **1004** and form conductive connections while establishing a physical (e.g., structural) attachment to the PCB **1002**. The reflow operation also causes the conductive connectors **832** to wet the conductive material of the interconnects **834** and **836** and form reflowed conductive connectors **1006** and **1008**. As a result, conductive connections between the semiconductor die **824** and the conductive pads **814** and **816** are formed interconnecting the packaged semiconductor die **822** with the radiating elements **806** and **808**.

(32) In this embodiment, the RF interposer further includes conductive vias **1010** interconnecting signal reflectors **902** and **904** with respective surrounding rings **810** and **812**. The conductive vias **1010** are configured and arranged to form a vertical conductive structure (e.g., wall, fence) substantially surrounding each of the respective radiating elements **806** and **808**. In this embodiment, the signal reflectors **902** and **904** are located at the top side of the substrate **804** opposite of the respective radiating elements **806** and **808** located at the bottom side of the substrate **804**.

(33) Generally, there is provided, a method including providing a radio frequency (RF) interposer, the RF interposer including a non-conductive substrate; a radiating element formed on the non-conductive substrate; and a cavity formed in the non-conductive substrate; affixing a packaged semiconductor die in the cavity of the RF interposer; and dispensing a conductive material to form a conductive path between a conductive connector of the packaged semiconductor die and the radiating element. The conductive connector may be formed as a ball including a solder material. The method may further include reflowing the solder material causing solder to wet a portion of the conductive path. The packaged semiconductor die may be affixed in the cavity by way of an adhesive material. The cavity may substantially surround the packaged semiconductor die on five sides. The radiating element may be formed on a top side of the non-conductive substrate and the cavity may be formed on a bottom side of the non-conductive substrate, a portion of the radiating

element overlapping the cavity. The RF interposer may further include a via configured to interconnect the radiating element with a conductive pad formed on the non-conductive substrate, the conductive path directly contacting the conductive pad. The radiating element may be formed on a bottom side of the non-conductive substrate and the cavity may be formed on the bottom side of the non-conductive substrate. The method may further include affixing a waveguide structure on the RF interposer, a waveguide formed in waveguide structure aligned over the radiating element.

(34) In another embodiment, there is provided, a method including providing a radio frequency (RF) interposer, the RF interposer including a non-conductive substrate; a radiating element formed on the non-conductive substrate; a conductive structure substantially surrounding the radiating element; and a cavity formed in the non-conductive substrate; affixing a packaged semiconductor die in the cavity of the RF interposer by way of an adhesive material; and dispensing a conductive path between a conductive connector of the packaged semiconductor die and the radiating element. The conductive connector may include a solder material. The method may further include reflowing the solder material causing solder to wet a portion of the conductive path. The RF interposer may further include a conductive layer portion formed on a side of the non-conductive substrate opposite from the radiating element, the conductive layer portion conductively connected to the conductive structure and configured to serve as a signal reflector. The dispensing a conductive path may include dispensing a conductive material to form the conductive path, the conductive material comprising a solder material. The radiating element may be formed on a top side of the non-conductive substrate and the cavity may be formed on a bottom side of the non-conductive substrate, a portion of the radiating element overlapping the cavity. The radiating element may be characterized as a signal launcher configured for propagation of an RF signal in a frequency range of 30 GHz to 300 GHz.

(35) In yet another embodiment, there is provided, a semiconductor device including a radio frequency (RF) interposer, the RF interposer including a non-conductive substrate; a radiating element formed on the non-conductive substrate; a conductive pad formed on the non-conductive substrate and interconnected to the radiating element; and a cavity formed in the non-conductive substrate; a packaged semiconductor die affixed in the cavity of the RF interposer by way of an adhesive material, the packaged semiconductor die having a plurality of conductive connectors; and a conductive material dispensed onto the conductive pad and a portion of the packaged semiconductor die adjacent to a conductive connector of the plurality of conductive connectors, the conductive material forming a conductive path between the conductive connector and the radiating element. The conductive connector may be reflowed with the conductive material to form the conductive path. The RF interposer may further include a conductive layer portion formed on a side of the non-conductive substrate opposite from the radiating element, the conductive layer portion configured to serve as a signal reflector. The cavity may substantially surround the packaged semiconductor die on five sides.

(36) By now, it should be appreciated that there has been provided a semiconductor device with an RF interposer. The semiconductor device includes a singulated packaged semiconductor die affixed in a cavity of the RF interposer. The packaged semiconductor die is formed as an encapsulated semiconductor die including a plurality of conductive connectors. The RF interposer includes a non-conductive substrate having radiating elements and other conductive and interconnecting features. The radiating elements may be configured for propagation of RF signals such as radar, Wi-Fi, UWB, 5G and 6G signals, for example. The cavity is formed in a bottom side of the substrate. With the packaged semiconductor die affixed in the cavity, conductive material is applied to form conductive interconnects between conductive pads of the RF interposer substrate and respective conductive connectors of the packaged semiconductor die. The conductive material may be applied by dispensing, jetting, and the like. A reflow operation causes the respective conductive connectors to wet the conductive material of the conductive interconnects forming conductive paths between the packaged semiconductor die and the radiating elements, for example. By forming the

conductive interconnects after the packaged semiconductor die and RF interposer are joined together, supply chain complexity and costs may be reduced.

(37) The terms “front,” “back,” “top,” “bottom,” “over,” “under” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(38) Although the invention is described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention. Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

(39) Furthermore, the terms “a” or “an,” as used herein, are defined as one or more than one. Also, the use of introductory phrases such as “at least one” and “one or more” in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an.” The same holds true for the use of definite articles.

(40) Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements.

Claims

1. A method of forming a semiconductor device, the method comprising: providing a radio frequency (RF) interposer, the RF interposer comprising: a non-conductive substrate with a bottom side; a conductive pad formed at the bottom side of the non-conductive substrate; a radiating element formed on the non-conductive substrate and interconnected with the conductive pad; and a cavity formed in the bottom side of the non-conductive substrate; affixing a packaged semiconductor die in the cavity of the RF interposer with an adhesive material that fills a space between sidewalls of the packaged semiconductor die and sidewalls of the cavity, wherein the packaged semiconductor die has a bottom side and a conductive connector at the bottom side of the packaged semiconductor die; after affixing the packaged semiconductor die in the cavity, dispensing a conductive material that extends from the conductive pad over the adhesive material in the space to the bottom side of the packaged semiconductor die to form a conductive path between the conductive connector at the bottom side of the packaged semiconductor die and the conductive pad at the bottom side of the non-conductive substrate; and reflowing to wet the conductive connector across the conductive material and interconnect the packaged semiconductor die with the radiating element.

2. The method of claim 1, wherein the conductive connector is formed as a ball comprising a solder material.

3. The method of claim 2, further comprising reflowing the solder material causing solder to wet a portion of the conductive path.

4. The method of claim 1, wherein the packaged semiconductor die also is affixed to a bottom surface of the cavity by way of the adhesive material, and the adhesive material is coplanar with a bottom surface of the semiconductor device.

5. The method of claim 1, wherein the cavity substantially surrounds the packaged semiconductor

die on five sides.

6. The method of claim 1, wherein the radiating element is formed on a top side of the non-conductive substrate and the cavity is formed on the bottom side of the non-conductive substrate, a portion of the radiating element overlapping the cavity.
7. The method of claim 1, wherein the RF interposer further comprises a via configured to interconnect the radiating element with the conductive pad formed on the non-conductive substrate, the conductive path directly contacting the conductive pad.
8. The method of claim 1, wherein the radiating element is formed on the bottom side of the non-conductive substrate and the cavity is formed on the bottom side of the non-conductive substrate.
9. The method of claim 1, further comprising affixing a waveguide structure on the RF interposer, a waveguide formed in waveguide structure aligned over the radiating element.
10. A method of forming a semiconductor device, the method comprising: providing a radio frequency (RF) interposer, the RF interposer comprising: a non-conductive substrate with a bottom side; a conductive pad formed at the bottom side of the non-conductive substrate; a radiating element formed on the non-conductive substrate and interconnected with the conductive pad; a conductive structure substantially surrounding the radiating element; and a cavity formed in the bottom side of the non-conductive substrate; affixing a packaged semiconductor die in the cavity of the RF interposer by way of an adhesive material that fills a space between sidewalls of the packaged semiconductor die and sidewalls of the cavity, wherein the packaged semiconductor die has a bottom side and a conductive connector at the bottom side of the packaged semiconductor die; dispensing a conductive path between the conductive connector at the bottom side of the packaged semiconductor die, over the adhesive material in the space, and to the conductive pad at the bottom side of the non-conductive substrate; and reflowing to wet the conductive connector across the conductive material and interconnect the packaged semiconductor die with the radiating element.
11. The method of claim 10, wherein the conductive connector comprises a solder material.
12. The method of claim 11, further comprising reflowing the solder material causing solder to wet a portion of the conductive path.
13. The method of claim 10, wherein the RF interposer further comprises a conductive layer portion formed on a side of the non-conductive substrate opposite from the radiating element, the conductive layer portion conductively connected to the conductive structure and configured to serve as a signal reflector.
14. The method of claim 10, wherein dispensing a conductive path includes dispensing a conductive material to form the conductive path, the conductive material comprising a solder material.
15. The method of claim 10, wherein the radiating element is formed on a top side of the non-conductive substrate and the cavity is formed on the bottom side of the non-conductive substrate, a portion of the radiating element overlapping the cavity.
16. The method of claim 10, wherein the radiating element is characterized as a signal launcher configured for propagation of an RF signal in a frequency range of 30 GHz to 300 GHz.
17. A semiconductor device comprising: a radio frequency (RF) interposer, the RF interposer comprising: a non-conductive substrate with a bottom side; a radiating element formed on the non-conductive substrate; a conductive pad formed on the bottom side of the non-conductive substrate and interconnected to the radiating element; and a cavity formed in the bottom side of the non-conductive substrate; a packaged semiconductor die affixed in the cavity of the RF interposer by way of an adhesive material that fills a space between sidewalls of the packaged semiconductor die and sidewalls of the cavity, the packaged semiconductor die having a plurality of conductive connectors formed at a bottom side of the packaged semiconductor die; a conductive material dispensed onto the conductive pad at the bottom side of the RF interposer, over the adhesive material in the space, and over a portion of a bottom side of an encapsulant of the packaged

semiconductor die adjacent to a conductive connector of the plurality of conductive connectors, the conductive material forming a conductive path between the conductive connector and the conductive pad; and a reflowed material wetted across the conductive material, the reflowed material interconnecting the packaged semiconductor die with the radiating element.

18. The semiconductor device of claim 17, wherein the reflowed material is solder wetted across the conductive material interconnecting the packaged semiconductor die with the radiating element.

19. The semiconductor device of claim 17, wherein the RF interposer further comprises a conductive layer portion formed on a side of the non-conductive substrate opposite from the radiating element, the conductive layer portion configured to serve as a signal reflector.

20. The semiconductor device of claim 17, wherein the cavity substantially surrounds the packaged semiconductor die on five sides.
